MASTER'S COURSE

Petrophysics

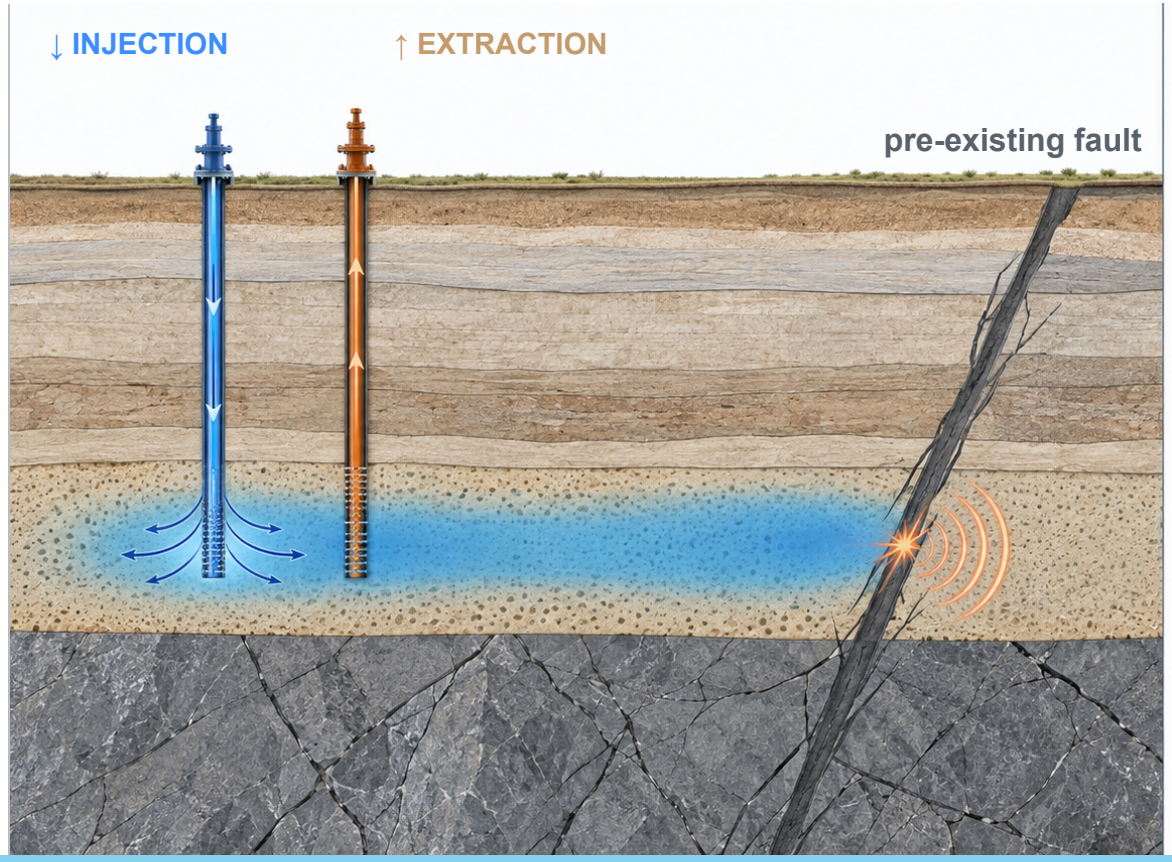
Experimental properties, deformation and fluid–rock interactions

LECTURE 1

How can we describe a rock quantitatively?

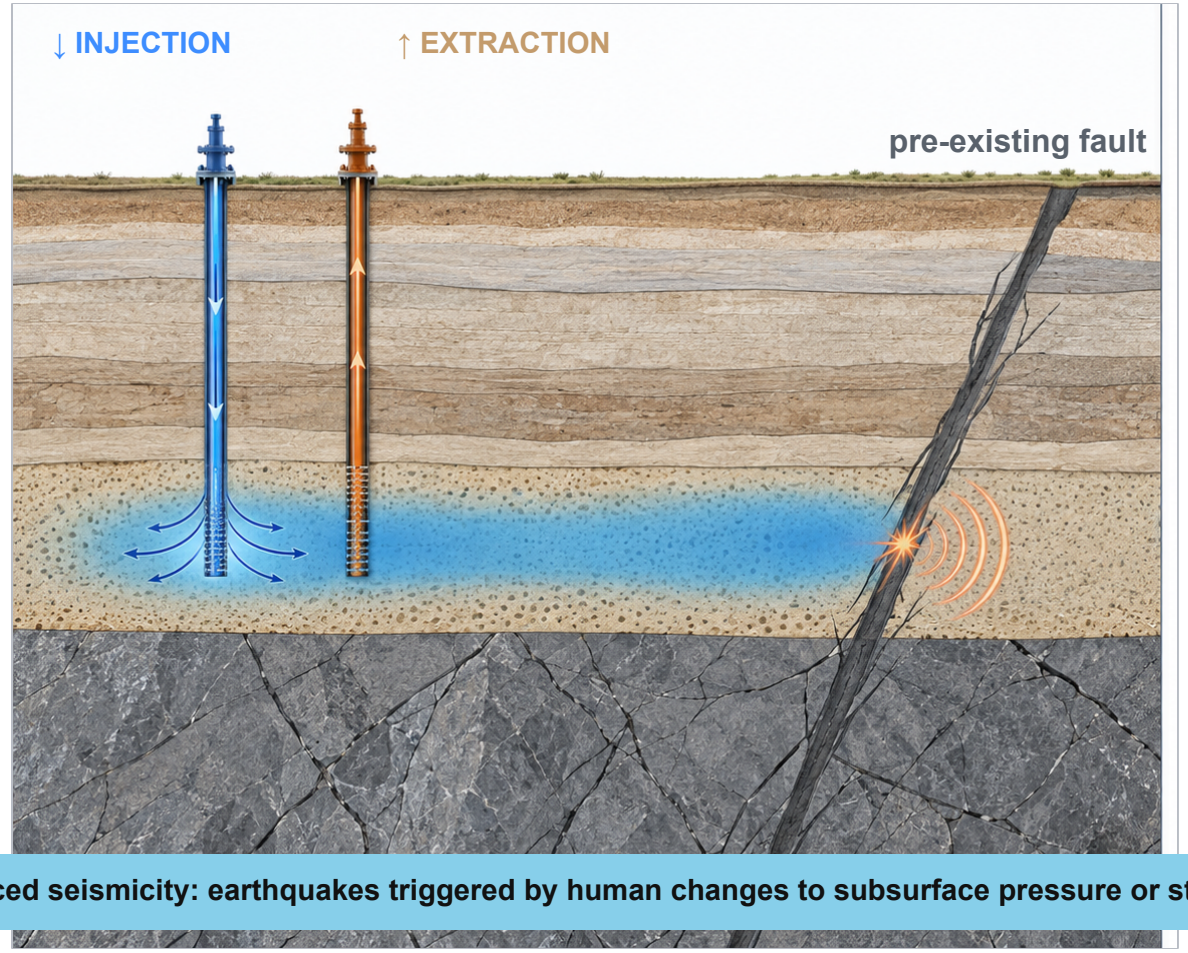
Prof. Marco Scuderi · Sapienza Università di Roma

Human fluid operations can trigger fault slip



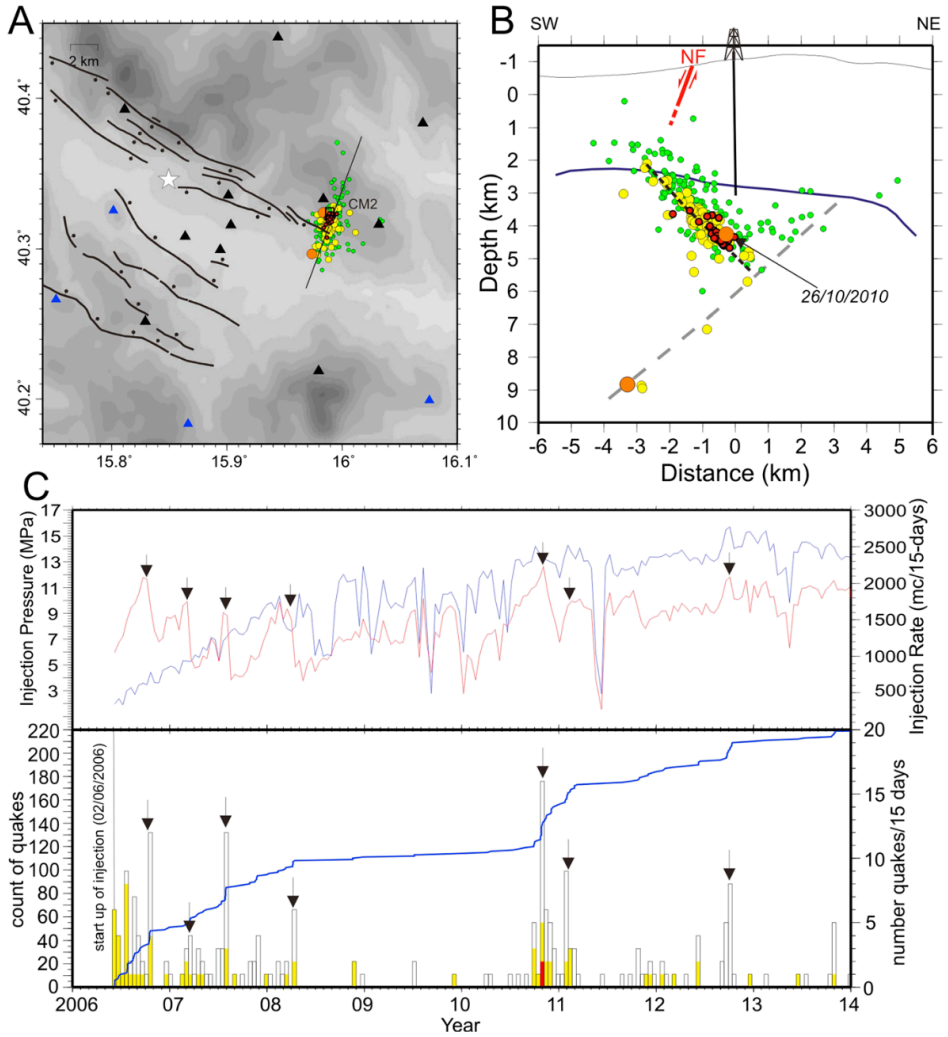
Induced seismicity: earthquakes triggered by human changes to subsurface pressure or stress.

Human fluid operations can trigger fault slip

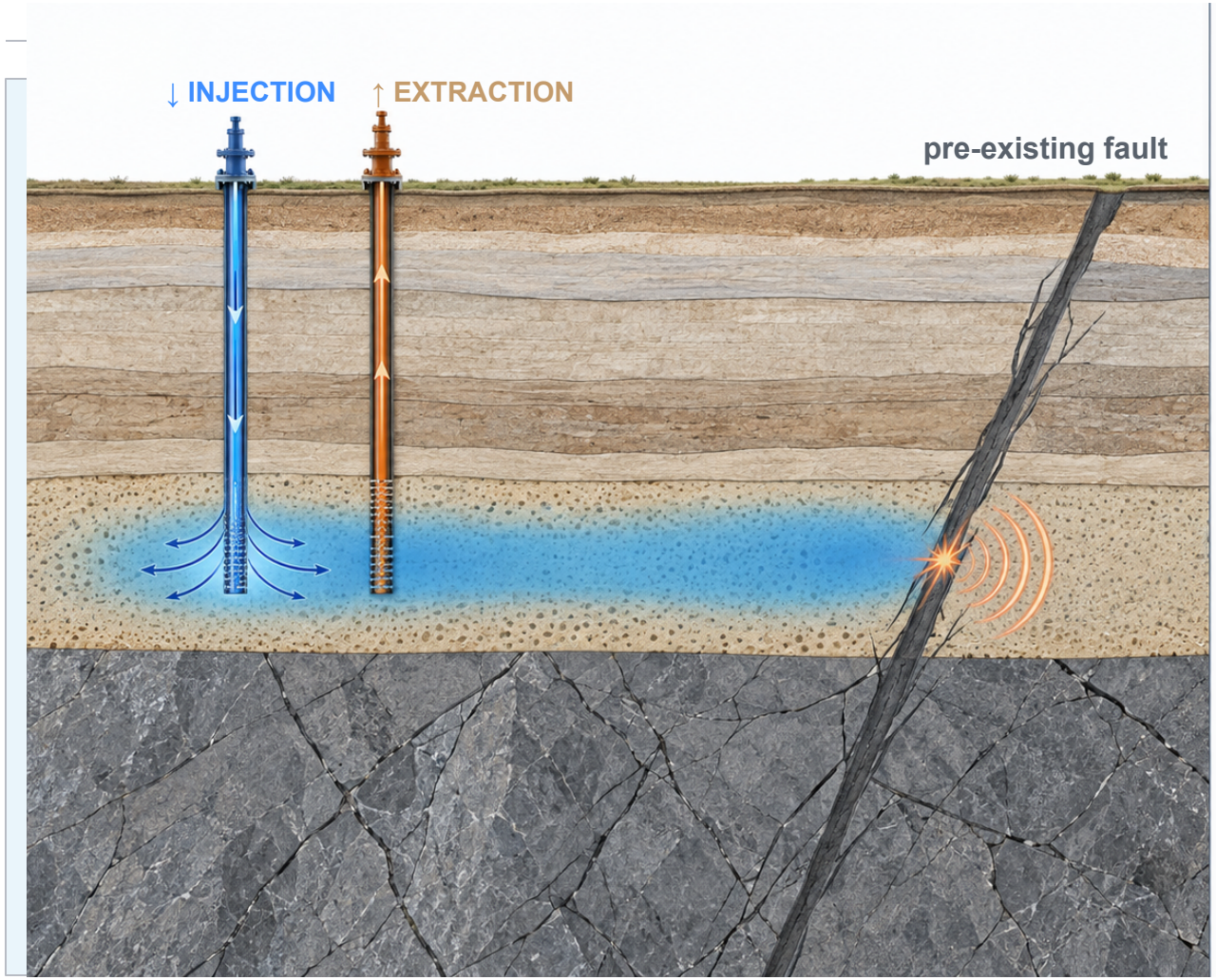


Induced seismicity: earthquakes triggered by human changes to subsurface pressure or stress.

Val D'Agri Oil field



Human fluid operations can trigger fault slip





ROCK AT THE INJECTION INTERVAL

Porous sandstone

connected grain pores

Fractured carbonate

pores and fractures

Fractured crystalline rock

fractures dominate

A small core may carry a very large decision

A subsurface formation is being evaluated for fluid injection.

Can this rock…

- store fluid?
- transmit it?
- deform safely?
- fracture or reactivate a fault?



THINK · PAIR · SHARE

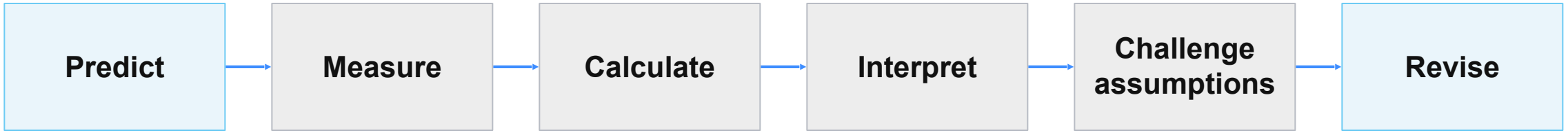
Before measuring, decide what matters

MEASUREMENTS	DECISION	LIMITATION
What would you measure?	What could the evidence inform?	How might this core mislead us?

2 min alone → 3 min with a partner

Petrophysics connects measurements to decisions

The course repeats one scientific cycle



Today we begin with what the rock is

Describe

bulk, solid and pore volume

Measure

density and water-accessible porosity

Infer

what fabric may explain

Critique

method, uncertainty and scale

Central question

With a scale, a caliper and water—what can we know?

PREDICTION

Look first: which rock contains the most void space?

bulk density | **porosity** | **permeability** | **P-wave velocity**

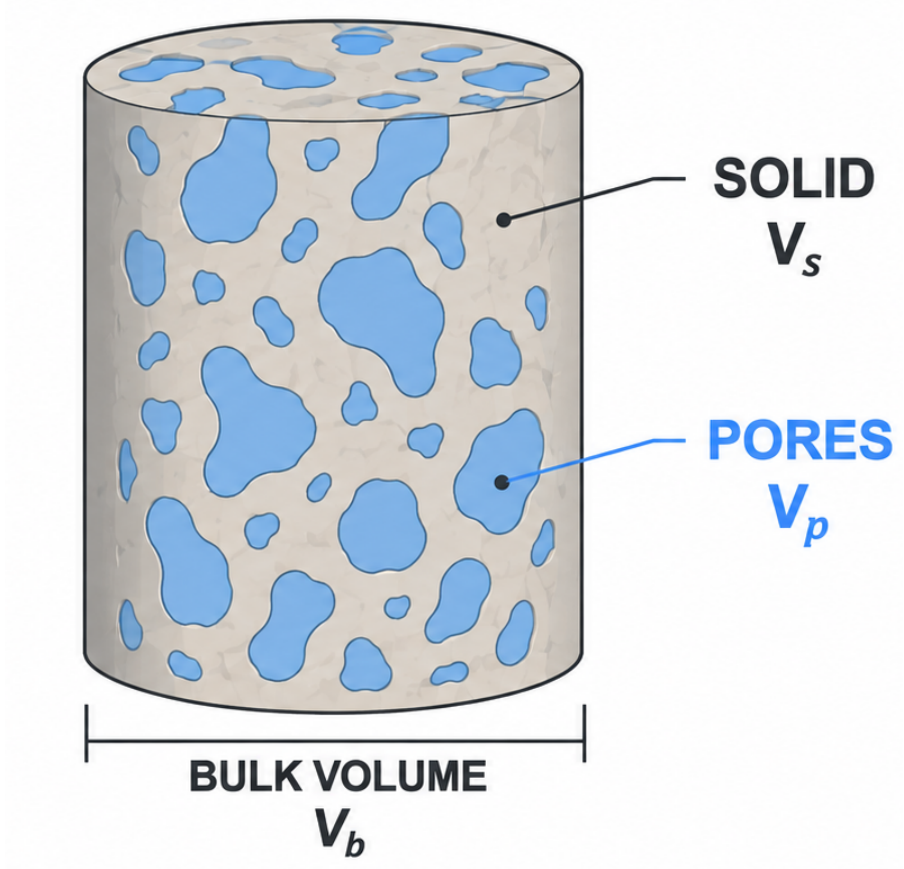
Rank the three specimens for each property.

Then mark your confidence: low · medium · high

Do not discuss yet.



Every rock sample contains solids and voids



$$V_\beta = V_s + V_p$$

The boundary of the specimen defines bulk volume.

Everything inside is assigned to solid or pore space.

CONCEPT CHECK

Different measurements see different volumes

Caliper dimensions	external envelope	V_β
Water uptake	water-accessible voids	$V_{p,e}$
Helium pycnometry	very small connected voids	**depends on method**

Question: where do closed pores go?

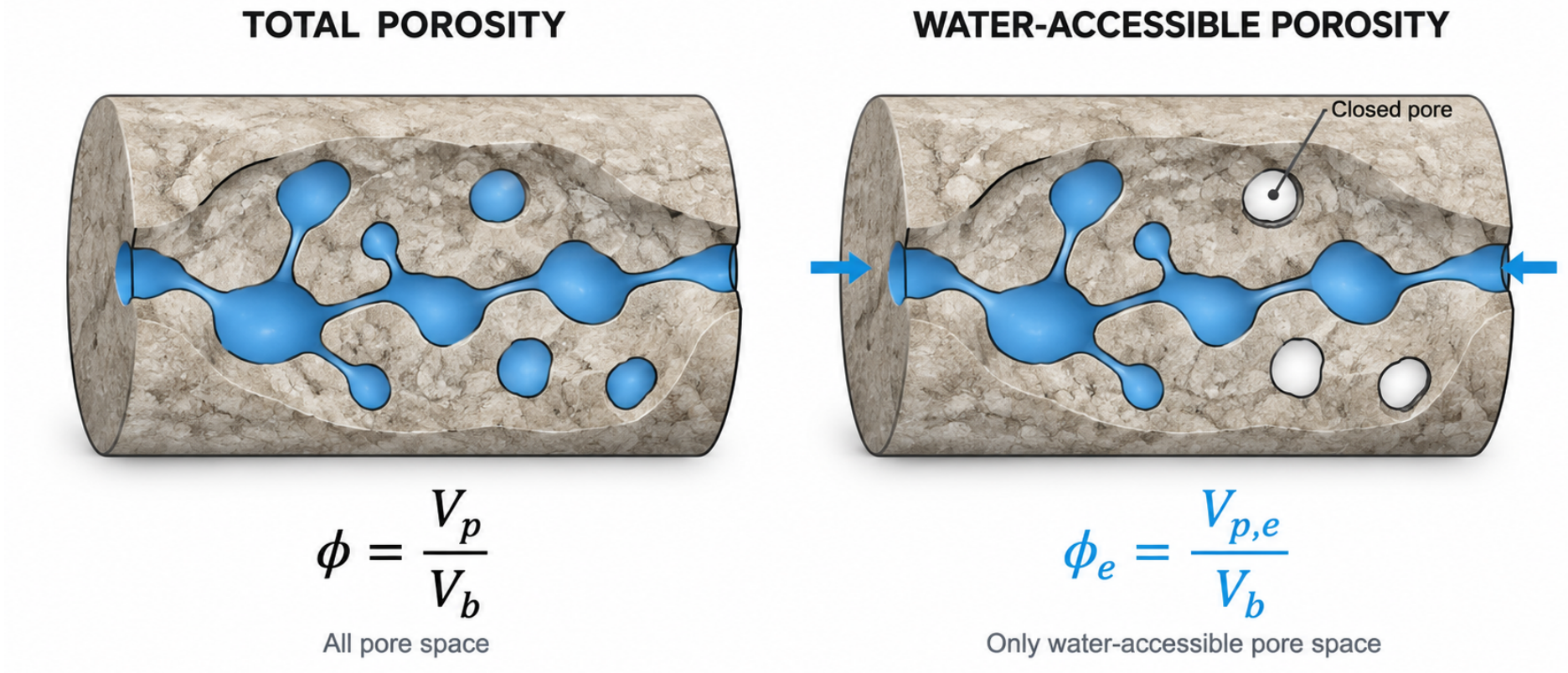
Density depends on the mass “state” and the reference volume

Dry bulk density	$\rho_{b,dry} = \frac{m_d}{V_b}$	solid mass over external volume
Saturated bulk density	$\rho_{b,\mathrm{sat}} = \frac{m_{\mathrm{sat}}}{V_b}$	solid + pore-fluid mass over external volume
Grain density	$\rho_g = \frac{m_d}{V_s}$	solid mass over solid volume

A density value is incomplete unless the mass state, reference volume and measurement method are specified

Think about: What is the role of stress in controlling density?

Porosity is a fraction—not a picture of the network



Neither value alone specifies pore shape, size, orientation or connectivity.